Q13 — Comprehension check

DLS 2026 — AI Internals Module

Knowledge representation, storage, and retrieval in transformers

A central puzzle in the study of large language models is *where* factual knowledge physically lives inside the network, and *how* it is surfaced at inference time. The dominant empirical account points to the feed-forward (MLP) sublayers as key–value memory stores: the first weight matrix acts as a bank of pattern-detecting "keys" that fire on specific input contexts, and the second matrix contributes corresponding "value" vectors that are written into the residual stream [3]. Causal interventions reinforce this picture: Meng et al. [5] show that a small set of mid-layer MLP modules are decisive for recalling a given subject–relation fact, and that editing those weights (ROME) can surgically alter the stored association. Complementary work attributes facts to neurons that, when ablated or amplified, change the predicted object [1]. Attention heads then play a retrieval role, moving information between token positions—induction heads, for instance, implement a copy-and-complete mechanism that underlies in-context recall [6]. The interpretability toolkit that connects these components treats the residual stream as a shared communication channel written and read by each layer [2], with intermediate "logit lens" readouts revealing how a prediction is progressively refined [4].

Question. A model reliably completes the prompt "*The Eiffel Tower is located in the city of*" with "*Paris*." (a) Using the key–value memory and causal-tracing perspectives, describe where this fact is most likely stored and by which mechanism it is retrieved and written to the output. (b) An engineer applies ROME to a mid-layer MLP so that the model now answers "*Rome*." Explain concretely what ROME changes and why editing that specific component suffices, and name one limitation such weight edits exhibit.

Answer. (a) The fact is stored predominantly in the second (output) weight matrix of one or more *mid-layer* MLP sublayers, operating as key–value memory [3]. As the subject token "Eiffel Tower" is processed, attention heads first aggregate the relevant subject information at the last token position—a routing step consistent with the induction/copy mechanisms and residual-stream communication framework [2, 6]. At a mid-layer MLP, a subset of keys in the first matrix detects this subject-plus-relation pattern ("located-in-city" context), and the paired value vectors in the second matrix are added into the residual stream, incrementing the component of the representation aligned with the "Paris" output direction. Causal tracing localizes this effect: corrupting the subject tokens and then restoring individual hidden states shows that restoring the mid-layer MLP outputs at the last subject token recovers the correct prediction, identifying those modules as the decisive site of storage [5]. The "logit lens" view shows the "Paris" prediction becoming readable in the residual stream after these layers write to it [4], and the final unembedding projects this to a high logit for "Paris."

(b) ROME treats the second MLP matrix W as a linear associative memory and performs a rank-one update $W' = W + \Lambda(C^{-1}k_*)^\top$ that inserts a new key–value association: it computes the key

vector k_* that the subject "Eiffel Tower" induces at the targeted layer, and chooses Λ so the retrieved value $W'k_*$ now points toward the "Rome" output direction while minimally disturbing other stored associations (hence the C^{-1} covariance-whitening term) [5]. Editing this one mid-layer MLP suffices precisely because causal tracing identified it as the pivotal read/write site for subject–relation facts; changing where the value is written at that bottleneck propagates to the final logits. A key limitation is that such weight edits generalize poorly and can damage unrelated knowledge: they may fail to update logically entailed or paraphrased queries, can be undone or confused by related edits, and risk collateral drift in nearby associations—so the edit is local and brittle rather than a robust belief update [1, 5].

References

[1] Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

[2] Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html.

[3] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

[4] Mor Geva, Avi Caciularu, Kevin Ro Wang, and Yoav Goldberg. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

[5] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[6] Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html.